

E-Commerce US Dataset Analysis

Report 4: Data Cleaning

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1. Scope

- Input: master_orders.csv (10,000 × 36) and items_enriched.csv (21,800 × 24) from the integration stage.
- Output: master_clean.csv (10,000 × 39) and items_clean.csv (21,800 × 23).
- Three categories of operations: type conversions, missing value treatment, and structural fixes.

2. Date/Timestamp Conversions

- All date columns in the source were stored as plain strings (Python object dtype).
- Conversion applied using pd.to_datetime() across 5 columns in master_orders and 1 in items_enriched.

Column	Table	Before	After
order_purchase_timestamp	master_orders	object (str)	datetime64[us]
order_approved_at	master_orders	object (str)	datetime64[us]
order_estimated_delivery_date	master_orders	object (str)	datetime64[us]
order_delivered_carrier_date	master_orders	object (str)	datetime64[us]
order_delivered_customer_date	master_orders	object (str)	datetime64[us]
shipping_limit_date	items_enriched	object (str)	datetime64[us]

3. Missing Value Treatment

3.1 Customer Age — Median Imputation

- 804 records (8.04%) had customer_age values outside the valid range [18, 100].
- Treatment: replaced invalid entries with the column median (48 years).
- Age range after treatment: 19 to 70 years.
- Median chosen over mean to avoid sensitivity to the invalid values themselves.

3.2 Delivery Date Nulls — Flag Encoding

- 668 rows had null order_delivered_carrier_date and order_delivered_customer_date.
- Root cause: all 668 are canceled orders; delivery dates structurally absent.
- Treatment: not imputed. Created binary flag is_delivered (1 = delivered, 0 = canceled).
- Delivery date nulls retained as-is; downstream analysis filters on is_delivered as needed.

3.3 Review Score Nulls — Flag Encoding

- 668 rows had null avg_review_score and n_reviews — same canceled-order records.
- Treatment: n_reviews filled with 0 for canceled orders; created has_review binary flag.
- avg_review_score remains null for canceled orders (no imputation — no review occurred).

4. Structural Fixes

4.1 Redundant Column Removal

- items_enriched contained a column price_prod that duplicated the price column from the products join.
- price_prod dropped from items_clean — no information loss.
- Column count: items_enriched 24 → items_clean 23.

4.2 Engineered Flags Added

Flag Column	Type	Logic	Use Case
is_delivered	Binary (0/1)	1 if order_delivered_customer_date is not null	Filter delivered vs. canceled
has_review	Binary (0/1)	1 if avg_review_score is not null	Filter reviewed orders
is_high_value	Binary (0/1)	1 if total_payment > \$3,857.70 (IQR upper	Identify premium orders

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5. Outlier Handling

- High-value orders (225 records, total_payment > \$3,857.70): retained and flagged with is_high_value = 1.
- Premium item prices (863 records in items, price > \$1,746): retained — represent valid high-end SKUs.
- Heavy products (2,297 records, weight > 8,135 g): retained — catalog includes large consumer goods.
- No records deleted from either table. All outliers are flagged or noted for downstream context.

6. Before vs. After Comparison

Attribute	Before Cleaning	After Cleaning
master_orders columns	36	39 (3 flags added)
items_enriched columns	24	23 (price_prod dropped)
Date column dtype	object (str)	datetime64[us]
customer_age range	Contains invalid values	[19, 70]
Null age records	804	0
Canceled order treatment	Nulls only	Nulls + is_delivered/has_review flags

7. Output Files

File	Rows	Columns	Location
master_clean.csv	10,000	39	data/processed/
items_clean.csv	21,800	23	data/processed/

- master_clean.csv is the primary input for all EDA and feature engineering steps.
- items_clean.csv is the primary input for item-level analysis and product-level feature building.

8. What Was Not Changed

- No records were deleted from either table.
- Outlier rows are retained — deletion would remove valid business data (premium orders, heavy goods).
- Categorical encoding (label encoding, one-hot) was not applied at this stage — deferred to feature engineering.
- No normalization or scaling applied — deferred to modeling preparation.